F18CE2 MV Calculus and Real Analysis B: Tutorial 1

1 Suppose that $|a-L|<r$, and $|b-L|<r$ where $r>0$. Show that

$$(a)\quad |2a+5b-7L|<7r \qquad (b)\quad |3a-2b-L|<5r.$$

2 Suppose that $r>0$ and $|a-x|<r,\ |b-y|<r$. Show that $|ab-xy|<(|x|+|b|)r$.

3 Suppose that $r>0$ and $\delta<\min\{1,r/9\}$. Prove that if $|x-4|<\delta$ then $|x^2-16|<r$.

4 Show that

$$\big||a|-|b|\big|\leqslant |a-b|, \quad \text{for any } a,\, b\in\mathbb{R}. \tag{1}$$

Hint: start with $|a|$ and use the adding and subtracting trick to get this looking like something involving $|a-b|$. Then do the same with $|b|$.

5 Let (a_n) be a sequence. Define what is meant by $\lim_{n\to\infty} a_n = L$.

Using this definition, prove that

$$(a)\ \lim_{n\to\infty}\left(5-\frac{3}{n}\right)=5\,, \qquad (b)\ \lim_{n\to\infty}\frac{n+4}{n+6}=1\,, \qquad (c)\ \lim_{n\to\infty}\frac{3\cos n}{n+1}=0.$$

6 Suppose that $\lim_{n\to\infty} a_n = L$. Prove directly from the definition of limit that

$$(a)\ \lim_{n\to\infty}(5a_n+3)=5L+3\,, \qquad (b)\ \lim_{n\to\infty}|a_n|=|L|$$

(in part (b), use (1)).

7 Give examples of sequences, one in each case of the following

(a) $\lim_{n\to\infty} a_n = 2$ and $a_n>2$ for all n.
(b) (a_n) bounded and (a_n) not convergent.
(c) (a_n) a convergent sequence, (b_n) a bounded sequence and (a_nb_n) is not convergent.
(d) (a_n) and (b_n) not convergent, yet $(3a_n+b_n)$ is convergent.
(e) (a_n) and (na_n) convergent, yet (n^2a_n) not convergent.
(f) (a_n) and (b_n) are not convergent, yet (a_nb_n) is convergent.
(g) (a_n) convergent and $a_n>0$ for all n, yet $\left(\frac{1}{a_n}\right)$ not convergent.

8 Suppose that $\lim_{n\to\infty} a_n = L$. Prove directly from the definition of limit that

$$(a)\quad \lim_{n\to\infty}(2a_{n+1}+3a_n)=5L \qquad (b)\quad \lim_{n\to\infty}(a_{n+2}-2a_{n+1}+a_n)=0.$$